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NPS/003 /004 – Technical Specification For Indoor 33kv Switchgear

1. Purpose

This document is the technical specification for indoor 33kV switchgear for use on the NEDL and YEDL networks.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DC/PRC/S/201 Part 1		33kV Circuit Breakers For Installation at Primary Substations and Supply Points
	Rev 4	Technical Specification: 33kV Circuit Breakers
		NEDL 33kV Switchgear Specification

2. Scope

This specification covers the technical requirements for 33kV circuit breakers for use on the NEDL and YEDL networks. It will be necessary to consider and include any project specific requirements as detailed in Appendix 3, Addendum To Supplier Requirements.

3. Technical Requirements

All circuit breakers shall comply fully with the requirements of the latest version of the Energy Network Association Technical Specification (ENATS) 41-36 and the latest versions of the standards and regulations to which 41-36 refers.

The circuit breakers will preferably have been assessed by the ENA Switchgear Assessment Panel (SAP) and have been awarded an ENA Notice of Conformity.

Appendix 1 of this document details the requirements selected within ENATS 41-36, or where the requirements of ENATS 41-36 have been supplemented. Note particularly the specified fault-break duties and X/R ratios.

Appendix 2 shall be completed for each type of circuit breaker unit.

Appendix 3 contains an Addendum to Supplier Requirements which lists any project specific requirements.

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4. References

4.1 External Documentation

Reference	Title
ENATS 41-36	Distribution Switchgear for Service up to 36kV (Cable and Overhead Conductor Connected)
BSI 381C 1996	Specification for colours for identification, coding and special purposes.

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K.Midgley	Standards Engineer	K.Midgley	14/02/2006

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G.Bolden	Purchasing Manager	G.Bolden	14/02/2006

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G. Gazda	Standards Team Co-ordinator	p.p. Dave Blackledge	14/02/2006

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S.Johnson	24/03/2006

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	20/02/2006

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	20/02/2006

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APPENDIX-1: Options Selected within ENATS 41-36

Particular of System	ENATS 41-36 Ref	
Voltage		33 kV
Frequency		50 Hz
Number of Phases		3
Neutral Earthing		Solid
Circuit Breaker Characteristics		
Number of poles		3
Class – indoor/outdoor	1.2	Indoor
Rated voltage	1.4.1	36 kV
Rated Insulation level. (lightning impulse voltage)	1.4.2	170 kV (195 kV across isolating distance)
Rated Frequency	1.4.3	50
Rated Normal Current Busbar TX incomer circuit breaker Bus Section circuit breaker Feeder circuit breaker 3 circuit development circuit breaker	1.4.4	2000A or 2500A 2000A or 2500A 2000A or 2500A 1250 1250
Rated Short-time withstand current	1.4.5	25 kA
Rated duration of short circuit	1.4.7	3 sec
Rated supply voltage of opening and closing devices and auxiliary and control circuits. a) Closing and tripping b) Indication c) Control	1.4.8	a) 110 V DC b) 110 V DC/AC (supply point) 110 V AC primary/3CD. c) 24V DC (telecontrol indication) 110 V DC
Rated supply frequency of closing and auxiliary circuits.	1.4.9	DC/50Hz
Rated short-circuit breaking current. Equal to rated short-time withstand current.	2.4.101	25 kA at an X/R of 37 (time constant of 120 ms) Methodologies in IEC 62271-100 and IEC 62271-308 give this an equivalent rating of 40kA at an X/R 14.1 (time constant of 45ms)
Rated short-circuit making current	2.4.103	2.7 x the tested fault break duty
Rated operating sequence.	2.4.104	O – 0.3s – CO – 15s – CO as required by ENATS 41-36 and also be capable of O – 10s – CO in order to comply with YEDL and NEDL autoreclose sequences.

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Rated capacitive switching currents : Rated cable-charging current.	2.4.107	50 A
Classification of mechanical operations M1 or M2	2.4.110	M2: 10000
Classification of electrical endurance – tested for autoreclose duty – not tested for autoreclose duty	2.4.111	Class E2 Autoreclose
Mechanism type	2.5.5 –2.5.7	Tenderer to state
Key interlock	2.5.11	
Gas monitoring	1.5.9	2 stage: 1st stage – Alarm. 2 nd stage – Trip inhibit + alarm
Voltage presence indicating system	1.5.19.1	
Insulation Medium		Non-oil.
Arc extinguishing medium		SF6/Vacuum
Colour of Paint		Dark admiralty grey 632 BS381C preferred
Power cable termination		1250A: up to 2 x 630sq mm / phase 2000A: 3 x 630 sq mm / phase 2500A: 4 x 630 sq mm / phase

CT Specifications¹

¹ The CT order should be from the table below (Top: nearest to busbar. Bottom: nearest to circuit)

Panel Type	CT Application	Continuous Current	Ratio
Transformer 132/33kV	Overall Biased Differential	2200A or 2750A	2000/1 V _k > (24R _{ct} +48) V _k > 60V I _e < 6mA @ (4R _{ct} +12)
	DOC	140%	2000/1A 5VA 5P10 CLASS 1 R _{ct} < 6.7 ohm
	LDC/Voltage Control	150%	1575/1A 15VA CLASS 3

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	Bus Zone CTs Discrim Zone	2200A or 2750A	1/1250T $V_k > (40R_{ct} + 118)$ $I_e < 10mA$ (20Rct +35) Rct < 3.2 ohm Class PX
	Bus Zone CTs Check Zone	2200A or 2750A	1/1250T $V_k > (40R_{ct} + 83)$ $V_k > 125V$. $I_e < 5mA @ (20 R_{ct} + 35)$. Rct< 3.2 ohm. Class PX
	Metering		To be specified if required
Bus Section			
	Bus Zone 1 Discrim.	2200A or 2750A	As on Tx Panel
	Bus Zone 2 Discrim.	2200A or 2750A	As on Tx Panel
Feeder: Solkor /OCIT	Unit Prot	1375A	1/600T $V_k > (42ct + 81)$. $I_e < 100mA @ V_k$. $I_e < 50mA @ 50V$. Class PX
	IDMT	140%	1000/1 5VA 5P20 CLASS 1
	Bus Zone Discrim	1375A	As on Tx Panel
	Bus Zone Check	1375A	As on Tx Panel
Feeder: Dist/OCIT	Distance Prot.	1375A	1/1250T $V_k > (22R_{ct} + 13)$ $I_e < 100mA @ V_k$. Class PX
	IDMT	140%	1000/1 5VA 5P20 CLASS 1
	Bus Zone Discrim	1375A	As on Tx Panel
	Bus Zone Check	1375A	As on Tx Panel
Feeder: BEF Inst/OCIT (Inc. Metering if required)	HSOC/BEF	1375A	1/1250T $V_k > (16R_{ct} + 64)$ $I_e < 20mA @ (2R_{ct} + 25)$ Rct < 9 ohm. Class PX
	OC/E	140%	1000/1 5VA 5P20 CLASS 1
	Bus zone CTs Discrim. Zone	1375A	As on Tx Panel
	Bus Zone CTs Check Zone	1375A	As on Tx Panel
	Metering 1 (If required)		To be specified if required
	Metering 2 (If required)		To be specified if required

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3 Circuit Development: Transformer Panel	BEF	1250A	1/1250T. $V_k > (6R_{ct} + 64)$. $I_e < 20\text{mA} @ (2R_{ct} + 25)$ $R_{ct} < 3.3 \text{ ohm}$ Class PX
	Unit Prot	1375A	1/1250T. $V_k > (42R_{ct} + 81)$. $I_e < 100\text{mA} @ V_k$ $I_e < 17\text{mA} @ 50\text{V}$. Class PX. (for upto 3 ct sets in parallel in this switchboard)
	Unit Prot	1375A	As above
3 Circuit Development: Y Panel (metered)	Unit Prot	1375A	1/600T. $V_k > (42R_{ct} + 81)$. $I_e < 100\text{mA} @ V_k$ $I_e < 15\text{mA} @ 50\text{V}$. Class PX. (for upto 3 ct sets in parallel in this switchboard)
			Other CTs as specified
	Metering		To be specified if required
3 Circuit Development: Y Panel	Unit Prot	1375A	1/1250T. $V_k > (42R_{ct} + 81)$. $I_e < 100\text{mA} @ V_k$ $I_e < 17\text{mA} @ 50\text{V}$. Class PX. (for upto 3 ct sets in parallel in this switchboard)
3 Circuit Development: X Panel	Unit Prot	1375A	1/1250T. $V_k > (42R_{ct} + 81)$. $I_e < 100\text{mA} @ V_k$ $I_e < 17\text{mA} @ 50\text{V}$. Class PX. (for up to 3 ct sets in parallel in this switchboard)
33kV Neutral CT	REF	2200A or 2750A	1/2000T $V_k > (24R_{ct}+48) > 60\text{V}$ $I_e < 6\text{mA} @ (4R_{ct}+12)$. Class PX
	SBEF	100% (500% for 3 secs)	1000/1A 2.5VA 5P5

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APPENDIX - 2: DECLARATION OF TECHNICAL PERFORMANCE TO BE COMPLETED BY THE MANUFACTURER

	MANUFACTURER			
	CIRCUIT BREAKER TYPE REFERENCE			
		ENATS 41-36 reference where applicable	UNIT	
	CIRCUIT BREAKER CHARACTERISTICS			
1	ENATS Certificate of Conformance			
2	Number of poles			
3	Class – indoor/outdoor	1.2		
	Type reference of vacuum bottles (if applicable)			
	Type reference of interrupter (if applicable)			
4	Rated voltage	1.4.1	kV	
5	Rated Insulation level. (lightning impulse voltage)	1.4.2	kV	
6	Rated Frequency	1.4.3	Hz	
7	Rated Normal Current Busbar Circuit breaker (TX incomer) Circuit breaker (Bus Section) Circuit breaker (Feeder) Circuit breaker (3 Circuit Development)	1.4.4	A	
8	Rated Short-time withstand current	1.4.5	kA	
9	Rated duration of short circuit	1.4.7	S	
10	Rated supply voltage of opening and closing devices and auxiliary and control circuits. Closing and tripping Indication Control	1.4.8	V	
11	Rated supply frequency of closing and auxiliary circuits.	1.4.9	Hz/DC	
12	Rated short-circuit breaking current	2.4.101	kA	
13	DC time constant.		msec	
14	Rated short-circuit making current	2.4.103	kA	
15	Rated operating sequence.	2.4.104		
16	Rated capacitive switching currents : Rated cable charging current.	2.4.107	A	
17	Classification of mechanical operations M1 or M2	2.4.110		

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18	Classification of electrical endurance – tested for autoreclose duty – not tested for autoreclose duty	2.4.111		
19	Mechanism type	2.5.5 –2.5.7		
20	Key interlock	2.5.11		
21	Gas monitoring devices.	1.5.9	Type	
22	Voltage presence indicating system (VPIS) fitted	1.5.19.1	Yes/No	
	VPIS: location of capacitor bushing			
23	Insulation Medium			
24	Arc extinguishing medium			
35	Colour of Paint			
	Power cable connection type			
26	Power cable: max number, size and type of cables/phase Transformer incomer Feeder			
27	Test bushing: Location of connection. Type of connection			
	OPERATING MECHANISM DETAILS			
28	Manufacturer and Type Reference of Operating Mechanism.			
29	Mechanism travel.		mm	
30	Closing Coils Number and type Type reference Power consumption		W	
31	Operating Coil-Open Number and type Type reference Power consumption		W	
32	Operating time – Close		msec	
33	Operating time – Open		msec	
34	Closing spring recharge time (motor driven); if applicable		sec	

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35	Spring charge motor: power consumption ; if applicable		W	
36	Closing actuator current; if applicable.		A	
37	Maximum dynamic floor loading		kN	
	SF6 GAS DETAILS			
38	Design pressure of gas system at 20°C Insulation Interrupters		Barg / MPa	
39	Operating pressure of pressure relief device		Barg / MPa	
40	Normal pressure at 20°C		Barg / MPa	
41	Close lockout pressure		Barg / MPa	
42	Trip (open) lockout pressure		Barg / MPa	
43	Minimum close pressure		Barg / MPa	
44	Minimum trip pressure		Barg / MPa	
45	Quantity of gas (mass and volume) per compartment Busbar Feeder Interrupters Total		Kg	
46	Method of monitoring pressure and temperature compensation			
47	Moisture content at commissioning (max.)		mg/kg	
48	Dew point temperature at commissioning (max.)		degC	
49	Maximum permissible acidity level (HF)			
50	Maximum permissible leak rate (<1% per annum)			
51	Mass of alumina/molecular sieve and type reference			
	DIMENSIONS AND WEIGHTS			

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52	Transformer incomer Height Width Depth Mass		mm mm mm kg	
53	Bus Section Height Width Depth Mass		mm mm mm kg	
54	Feeder Height Width Depth Mass		mm mm mm kg	
55	Minimum clearances required to achieve internal arc performance from: Rear Side Ceiling Front		mm mm mm mm	

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APPENDIX-3: Addendum to Supplier Requirements

Project specific installation and protection requirements will be provided by Primary Engineering Projects for inclusion in this appendix.